



DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Experiment No.:3

Write a program to recognize a document is positive or negative based on polarity words using suitable classification method.

Date of Performance:

Particulars	Marks
Regularity(05)	
Performance(15)	
Understanding(Viva)(05)	
Total (25)	
Signature of Course Teacher	



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Aim:

- To develop a program that recognizes whether a document is positive or negative based on polarity words using a Naïve Bayes classification methods.

Objective:

- To understand sentiment analysis.
- To identify positive and negative polarity words in a document.
- To classify text using a machine learning classification method.

Theory:

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine whether a piece of text expresses positive, negative, or neutral sentiment.

A polarity word indicates emotional orientation:

- Positive words: good, excellent, happy, amazing, great
- Negative words: bad, poor, terrible, sad, worst

In this experiment we use the Naïve Bayes Classifier, a probabilistic machine learning algorithm based on Bayes' Theorem. It calculates the probability that a document belongs to a particular class (positive or negative).

Advantages of Naïve Bayes:

- Simple and fast
- Works well for text classification
- Requires less training data

Algorithm:

1. Import required libraries such as pandas and scikit-learn for text processing and classification.
2. Create a dataset that contains text reviews or Twitter comments and their sentiment labels (positive or negative).
3. Convert text data into numerical form using CountVectorizer so that the machine learning model can process the words.
4. Split the dataset into two parts:

Training data – used to train the model.

Testing data – used to check the model accuracy.



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5. Train the classification model using the Naïve Bayes algorithm.

6. Give a new Twitter comment as input to the trained model.

Example Input:

"I really love this new update, it is amazing!"

7. Predict the sentiment of the input comment.

8. Display the result showing whether the document is Positive or Negative.

Conclusion:

Sentiment classification helps in understanding public opinion from text data such as reviews, feedback, and social media posts. Naïve Bayes provides a simple and efficient approach for document sentiment classification.



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Experiment No.:4

Download the **Abalone Dataset** and build a model to predict the **number of rings (age)** of abalone using physical measurements with **Linear Regression**.

Date of Performance:

Particulars	Marks
Regularity(05)	
Performance(15)	
Understanding(Viva)(05)	
Total (25)	
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Aim:

To download the **Abalone Dataset** and build a model to predict the **number of rings (age)** of abalone using physical measurements with **Linear Regression**.

Problem Statement:

Predict the number of rings of an abalone using its physical measurements.

The number of rings indicates the age of the abalone.

Age calculation:

$$\text{Age} = \text{Rings} + 1.5$$

The dataset contains 4177 instances and 8 features such as length, diameter, height, and weight measurements.

Objective:

- Download the dataset from the UCI repository.
- Predict the number of rings using regression.
- Estimate the age of abalone using Linear Regression.
- Evaluate the prediction accuracy.

Algorithm:

1. Import Required Libraries

Import the necessary Python libraries for data processing and machine learning.

2.Download and Load the Dataset

This step loads the dataset into a table format called a DataFrame so that we can analyze and process it.

3.Convert Categorical Feature (Sex) into Numeric Values

The Sex column contains categorical values:

- M = Male
- F = Female
- I = Infant

Machine learning models work with numbers, so we convert these into numeric form



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4. Split Dataset into Training and Testing Data.

The dataset is divided into two parts:

- **Training Data** (80%) → used to train the model
- **Testing Data** (20%) → used to evaluate the model

5. Train the Linear Regression Model

The model learns the relationship between physical measurements and rings.

6. Predict the Number of Rings

After training, the model predicts the number of rings for test data.

7. Evaluate Model Performance

Mean Squared Error (MSE) measures prediction error.

Lower MSE = Better model performance.

Theory:

The Abalone Dataset is commonly used in machine learning for regression analysis. The dataset contains physical measurements of abalone (a type of marine snail). The main goal is to predict the age of an abalone based on its physical features.

Traditionally, the age of abalone is determined by counting the number of rings in the shell under a microscope. However, this process is time-consuming and requires expertise. Therefore, machine learning techniques are used to estimate the age using easily measurable physical attributes.

The dataset includes 4177 instances and several attributes such as:

- Length
- Diameter
- Height
- Whole weight
- Shucked weight
- Viscera weight
- Shell weight
- Sex (Male, Female, Infant)



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- Rings (target variable)

The number of rings indicates the age of the abalone.

Age of Abalone = Rings + 1.5

The additional 1.5 years accounts for the time before the shell starts forming rings.

Conclusion:

In this experiment, we applied Linear Regression to estimate the age of abalone from physical measurements. The model learns the relationship between shell measurements and the number of rings, enabling age prediction without manual inspection.



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Experiment No.:5

Apply the **K-Means Clustering** algorithm to a dataset containing eight 2-dimensional

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Regularity(05)	
Performance(15)	
Understanding(Viva)(05)	
Total (25)	
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Aim:

To implement the **K-Means Clustering** algorithm to group the given data points into clusters and update the cluster centroids.

Objectives:

- To understand the concept of clustering in machine learning.
- To apply the K-Means algorithm for grouping similar data points.
- To calculate the distance between data points and centroids.
- To update cluster centroids after assigning points to clusters.

Theory:

Clustering is an unsupervised learning technique used to group similar data points together. One of the most popular clustering algorithms is K-Means Clustering.

In K-Means clustering, the dataset is divided into **K clusters**, where each cluster is represented by a centroid (mean point). The algorithm works by assigning each data point to the nearest centroid and then recalculating the centroid based on the points assigned to that cluster.

The distance between the data point and centroid is usually calculated using Euclidean distance.

The K-Means algorithm repeats the following steps until the centroids stop changing:

- Assign data points to the nearest centroid.
- Recalculate the centroid by taking the mean of the points in the cluster.

This method helps in identifying patterns or groups within the dataset.

Algorithm

1. Select the Number of Clusters (K)

Choose how many groups you want in the dataset.

Example:

$$K = 2$$

So the dataset will be divided into **two clusters (C1 and C2)**.

2. Choose Initial Centroids

Select initial center points for each cluster.

Example: